

S8 – Clause Mapping Analysis

S8 Sensitivity Analysis – Clause Mapping

Purpose

The purpose of this analysis is to demonstrate how specific regulatory clauses from NHS Digital DCB0129 and related standards map to the three-level “Regulatory Embedding” dimension used within the proposed taxonomy. Furthermore, this analysis evaluates the robustness of the mapping process against variations in expert interpretation.

Method

For each of the six taxonomy dimensions, the Delphi expert panel identified the regulatory clauses considered most relevant to cybersecurity risk assessment and model governance. Two Clinical Safety Officers independently reviewed the resulting mappings with experience in NHS Digital compliance and healthcare safety assurance. Any discrepancies identified during the review process were subsequently resolved through structured discussion and consensus agreement.

Clause Mapping Table

DCB0129 Clause / Related Standard	Regulatory Embedding Level	Justification
Clinical Risk Management (DCB0129 §4.1)	Explicit Mapping	Requires documented risk analysis directly linked to model accuracy, explainability, and edge readiness requirements.
Software Development Lifecycle (DCB0129 §5.2)	Clause-Level Enforcement	Mandates traceability from requirements through verification and validation, aligning strongly with compliance and explainability dimensions.
Clinical Safety Case (DCB0160 §6)	Explicit Mapping	Requires a structured safety argument; mapping to compliance and accuracy dimensions is implicit yet substantively strong.
Information Governance (DCB0129 §7)	Nominal Mention	Provides general data protection requirements; however, the relationship to specific machine learning metrics is indirect.
Network Security (ISO 27001 A.13)	Explicit Mapping	Network segmentation and monitoring directly support edge readiness, resilience, and latency requirements.
Incident Management (DCB0129 §9)	Nominal Mention	Supports post-market surveillance and operational maintenance, although not directly measurable through model-level metrics.

Sensitivity Check

To evaluate mapping robustness, the two independent reviewers additionally completed the mapping exercise blind to the original assignments. The resulting inter-rater agreement achieved a Cohen’s kappa coefficient of $\kappa = 0.89$, indicating almost perfect agreement.

Subsequently, ambiguous clauses were intentionally reclassified to the next higher regulatory embedding level to assess interpretational sensitivity. Under these conditions, taxonomy scores across all evaluated models changed by ≤ 0.02 on a normalised 0–1 scale, demonstrating that the mapping framework remains highly stable under plausible interpretational variation.

Conclusion

The clause mapping analysis demonstrates that the regulatory embedding framework is both theoretically grounded and operationally robust. The strong inter-rater agreement and minimal score variation under reclassification scenarios indicate that the mapping process is resilient to reasonable expert disagreement. Consequently, the mapping structure provides a defensible basis for the compliance and regulatory embedding dimensions incorporated within the taxonomy.